Pakhi Banchalia

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Education

Technische Universität Dresden 2025 – Present
MSc in Computational Modeling and Simulation, Applied AI track (3rd Semester)

Amrita Vishwa Vidyapeetham, Amritapuri 2021 – 2025
Bachelor of Technology in Computer Engineering, Honors in Artificial Intelligence

Experience

Summer Research Intern, Aalto University Jun 2026 – Aug 2026
Espoo, Finland
- Selected as one of **67 interns from 5,500 applications** across 133 countries for the AScI International Summer Research Programme, working on **empowerment-based assistance in multi-agent reinforcement learning**.
- Found that AvE (NeurIPS 2020) tests assistance with an agent acting remotely rather than embodied; rebuilt the experiment with an embodied assistant to measure what that costs.

Research Intern, Lossfunk Jul 2025 – Sep 2025
Bangalore, India
- Built and evaluated **homeostatic RL agents** in **Crafter** driven by hunger, energy and safety, and explored intrinsic drives such as **curiosity** to support exploration alongside survival.
- Tested how reward formulations and competing drives affect exploration, survival, behavioural diversity and policy stability, against standard RL baselines.

Research Intern (Independent Collaboration), CMC Lab, TU Dresden Nov 2024 – Sep 2026
Dresden, Germany (Intermittent)
- Designed RL agents with **internal physiological drives** to study retention and generalization across changing task distributions.
- Analysed neural and behavioural data from trained foraging agents, decoding movement and internal state from network activity and linking oscillating unit groups to gait.

Contributor, Google Summer of Code (INCF) GitHub May 2024 – Sep 2024
Remote
- Built a **growing graph Neural Cellular Automaton** of the *C. elegans* embryo that adds each cell as it divides and learns where daughter cells sit in 3D, trained with **CMA-ES evolutionary optimization**.
- Mapped cell-by-cell embryo data with **Topological Data Analysis** (Kepler Mapper) into interactive visualizations, released as documented open-source code.

Core Committee Member; Design Lead, amFOSS Sep 2022 – Jul 2025
Amritapuri, India
- Led **amFOSS**, a **student-run open-source software community**, as a core committee member: ran weekly project check-ins, caught stalled work, and was the link between the founder, university staff and members.
- Headed the **design team**, reviewing work and making final design decisions, including the **amfoss.in** redesign.

Projects

Polaris: Rare-Disease Atlas Live GitHub 2026
- Built a rare-disease knowledge graph at the **Hack-Nation Global AI Hackathon** (Buffalo Initiative and OpenAI challenge), mapping patient communities by mechanism and mutation direction, every link sourced.
- Co-developed the idea, then built the interface and researcher view, which screens 2,868 approved medicines by mutation direction.

Alignd: AI Sponsorship Agent GitHub 2026
- Selected as **one of 60 demos from 2,000+ applications** to the TinyFish Accelerator, a $2M program run by TinyFish and Mango Capital, and reached **Phase 2**.
- Built an AI agent that finds and vets YouTube creators for a brand's campaign in six stages, deciding what the agent may conclude alone and what the brand can overrule.

MedX: X-ray Editing Tool 2024

- Reached the **final round of GE HealthCare's Precision Care Challenge 2024** with a web tool that edits X-ray images to generate training data.
- Owned the Next.js and TypeScript front end: visual design, layout, and the canvas where a user places an object into an X-ray.

Hospital Inventory System 2022 – 2023

- Deployed a Django inventory-management system for Amrita Hospital as part of a team, built to the hospital's requirements.
- Focused on building the ticketing and activity workflow for creating, assigning, tracking and resolving issues, plus UI work and bug fixes.

Publication

CAN: Continuously Adapting Network [Springer, ICDSA 2025] Link 2025

- Proposed a **continual-learning network** that learns new tasks without forgetting old ones by freezing task-critical neurons and applying **Hebbian learning**.
- Extended CAN in my B.Tech. thesis, **SEANN (Self-Expanding and Adaptive Neural Networks)**, so the network **grows its own capacity** and finds task-relevant subnetworks.

Talks and Posters

Emergence of Brain-like Features in Homeostatic RL Agents Link 2025

- Accepted as an **oral talk at BrainBody 2025**; co-authored work showing how internal states, neural dynamics and adaptive exploration are coupled.

Brain-like Features in Homeostatic Deep RL Agents Link 2025

- Accepted as a **poster at Bernstein Conference 2025**; co-authored work showing that agents combining internal and sensory inputs develop structured behavioural and neural patterns.

Skills

Programming Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, MATLAB, HTML, CSS
Machine Learning and AI: PyTorch, TorchVision, Gymnasium, JAX, Reinforcement Learning, Multi-Agent Reinforcement Learning, Continual Learning, Graph Neural Networks, Neural Cellular Automata, AI Agents, LLM APIs, Anthropic API, Gemini API
Data Science: NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper, Weights and Biases
Software Development and Design: FastAPI, Django, Next.js, React, n8n, Git, Linux, Docker, Figma
Languages: English (C1); German (A2)